

MOHAMMED AMIR KHAHAZA

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EDUCATION

Georgia Tech — *M.S. in Cybersecurity - (Information Security) - Online*

August 2026 - Present

UC Merced — *B.S. in Computer Science & Engineering*

Graduated: May 2026

GPA: 3.83

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Machine Learning, Discrete Math, Boolean Algebra

SKILLS

Programming: Python, C++, Java, JavaScript/TypeScript, SQL, Ruby, Go, MATLAB, Kotlin

AI & Data Tools: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, MongoDB, Tableau

Systems & Platforms: React, Node.js, Flask, FastAPI, Git, Perforce, Linux, CUDA, OpenCV, MediaPipe

Specialties: Hardware-in-the-Loop Integration, Motion Control Telemetry, Web Automation, Data Analysis

EXPERIENCE

Transportation Research Center Inc. (Waymo Test Services) - East Liberty, OH

Test Engineer I

August 2026 - Present

- Execute comprehensive **test plans** and complex driving scenarios to **validate the performance and safety of autonomous vehicle hardware and software systems**.
- Monitor **in-vehicle systems** and **sensor arrays** during live testing, meticulously documenting anomalies, **edge cases**, and **software bugs**.

NASA Ames Research Center – Mountain View, CA

Software Engineer Intern

May 2026 – August 2026

- Developed a Python pipeline (**NumPy, Pandas**) to synchronize hardware clocks and correct physical tracking drift using **2D affine transformations**, establishing highly accurate testing parameters for **motion tracking**.
- Engineered a cross-platform **PsychoPy** frontend for **9-point eye-tracker calibrations**, logging **VSYNC-aligned timestamps** for precise **gaze telemetry**.
- Computed **kinematic** features and integrated **LLMs** to analyze oculomotor behavior and mental workload for a lunar-lander analog tracking study.

PADSYS – Merced, CA

Undergraduate Research Intern

August 2025 – May 2026

- Investigating I/O bottlenecks in High-Performance Computing (HPC) systems by profiling AI workloads, focusing on processes like **checkpointing** and **data loading**.
- Deploying and benchmarking computer vision models (**YOLOv12, mmdetection library**) in containerized Ubuntu environments using **Docker** to analyze **I/O performance**.

PROJECTS

AIView: Control with Vision (Winner, SF Hacks 2025)

Team Lead & Developer

March 2025 - Present

- Engineered a **hands-free computer control system** for people with disabilities using **eye-tracking, blink detection, and speech commands**.
- Implemented computer vision with **Python, OpenCV, MediaPipe**, and Kalman filters; integrated **speech-to-text AI** and **MongoDB Atlas**.
- Improved gaze accuracy by 32% and reduced latency by 18% through adaptive AI filtering.

Devpost: <https://devpost.com/software/aivue-control-with-vision>

EthicScope (Winner, HackDavis 2025)

Founding Developer

April 2025 - May 2026

- Built a barcode-scanning web and mobile app using ML to assess **ethical** and **environmental sustainability**.
- Built backend with **Flask, MongoDB, and Cerebras**; developed visual data dashboards using **React + Tailwind**.
- Helped over 1,000 simulated test users make informed, transparent purchasing decisions.

Devpost: <https://devpost.com/software/ethicscope>